

Luca Mouchel

Visiting Student at MIT | Msc. Student in Data Science at EPFL

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Education

École Polytechnique Fédérale de Lausanne, MSc. in Data Science Sept. 2023 – August 2026

- Specializing in Data Analysis, AI, NLP, Reinforcement Learning and Mathematics in Data Science.
- One of 15 recipients of the MSc Research Scholarship, working part-time at the AI Lab for 1.5 years.

École Polytechnique Fédérale de Lausanne, BSc. in Communication Systems Sept. 2020 – June 2023

- Relevant Coursework: Signal Processing, Stochastic Models, ML, Probability and Statistics, Algorithms, Software Engineering.

Experience

Visiting Researcher, MIT – Boston, MA Feb. 2026 – Aug. 2026

- Redefining occupational AI exposure by developing a RAG pipeline with news articles and research papers with open source models using vLLM. Featured in the Boston Globe - [Click to view article](#).

Machine Learning Research Intern, SLB-Research – Boston, MA July 2025 – Jan. 2026

- Integrated and patented Vision Transformers (ViT) into the production training pipeline for sonic log analysis. Developed a highly modular architecture trained from scratch, making the model components modular. Implemented AutoKeras for automated hyperparameter optimization, streamlining model tuning and improving training efficiency.
- Developed methods to visualize and interpret how CNN models process sonic image data derived from slowness-time coherence wavelet transforms with GradCam++

Research Assistant – *Apertus* contributor, SwissAI Initiative – Lausanne, Switzerland Jan. 2025 - July 2025

- Contributed to Apertus's post-training. Initialized online reinforcement learning infrastructure using the VERL framework.
- Developed a novel curriculum learning algorithm on top using Reasoning-Gym, to improve RL training efficiency and stability. Scaled the implementation to 70B-parameter models using distributed Ray clusters, handling the computational complexity of large-scale online RL with systematic difficulty progression throughout training.

Research Scholar, AI Lab, EPFL – Lausanne, Switzerland Sept. 2023 - Feb. 2025

- Conducted research part-time on **reasoning** in LLMs as one of the Research Scholarship recipients. Led several projects, including improving LLMs' capabilities at generating logically sound arguments with preference optimization.
- Co-authored two other papers: how LLMs can maintain epistemic consistency and a survey paper on Uncertainty in Causality.

Research Assistant, ML for Education Lab, EPFL – Lausanne, Switzerland Sept. 2022 - Feb. 2023

- Built a data analysis pipeline, combining data mining and NLP, to analyse self-regulated learning behaviors in keystroke logs.

Achievements

- Contributed to the post-training experiments of the Swiss foundation model *Apertus* (2M+ Downloads on HuggingFace)
- *Outstanding Paper Award*: "A Logical Fallacy-Informed Framework for Argument Generation" received the award at NAACL 2025.
- *Research Scholarship* (2023-2025): One of 15 research scholarship recipients for 2023/24 to work part-time in a lab.
- On the *Program Committee* for AAAI 26', ACL 26' and Neurips 26'
- *Publications at Master's Level*: 6 Publications at top venues including AAAI, ACL, NAACL, Springer Nature) with 80+ citations.
- *Travel Scholarships*: Received travel scholarships to travel to the EDM Conference in India and to NAACL in the US.

Selected Publications

- *Apertus*: Democratizing Open And Compliant LLMs For Global Language Environments
- **Luca Mouchel**, Debjit Paul, Shaobo Cui, Robert West, Antoine Bosselut, Boi Faltings. A Logical Fallacy-Informed Framework for Argument Generation. *NAACL, 2025, (Outstanding Paper Award)*.
- Shaobo Cui, **Luca Mouchel**, Boi Faltings. Uncertainty in Causality: A New Frontier. *ACL, 2025*.
- **Luca Mouchel**, Thimo Wamborganss, Paola Mejia, Tanja Käser. Understanding Revision Behavior in Adaptive Writing Support Systems. *International Conference on Educational Data Mining, 2023*
- Shaobo Cui, Junyou Li, **Luca Mouchel**, Yiyang Feng, Boi Faltings. Nuance Matters: Probing Epistemic Consistency in Causal Reasoning. *AAAI, 2025*

Technologies & Skills

Languages: English, French, Italian, Spanish | **Programming Languages:** Python, Java, Scala, C, SQL

Technologies: Git, HuggingFace, Vim, Docker, RunAI, SLURM, PyTorch, Tensorflow, Keras, vLLM